

February 16, 2016

Hon. Maria Pallante
Register of Copyrights
United States Copyright Office
101 Independence Ave., S.E.
Washington, D.C. 20559

Re: **Software-Enabled Consumer Products Study; Notice and Request for Public Comment**
Docket No. 2015-6

Dear Ms. Pallante:

Microsoft welcomes this opportunity to respond to the Copyright Office's Notice of Inquiry for its Software-Enabled Consumer Products Study ("Notice").¹ As Senators Grassley and Leahy noted in their letter to the Copyright Office of October 22, 2015, the ubiquity of innovative software in everyday consumer products such as cars, refrigerators, and other devices provides tremendous benefits to consumers, but also raises important policy issues.² Consistent with their letter, and with the terms of the Notice, this submission focuses on the role of copyright protection for software embedded in consumer products; it does not address other policy issues raised by such products (*e.g.*, privacy or security), or the well-settled copyright eligibility of all other types of software.

These comments are informed by more than three decades of Microsoft leadership in nearly all facets of software innovation. In addition to developing some of the world's most popular software products, Microsoft also offers several leading platforms, services, tools, and devices that allow customers to run and adapt third-party software, or even develop software themselves. The broad diversity of our offerings gives us a unique—and, we believe, uniquely balanced—perspective on the issues raised in the Notice.

¹ 80 Fed. Reg. 77668 (Dec. 15, 2015).

² Letter from Sens. Charles E. Grassley and Patrick Leahy to Hon. Maria A. Pallante (Oct. 22, 2015), at <http://www.grassley.senate.gov/news/news-releases/grassley-leahy-call-copyright-study> (last accessed Feb. 12, 2016).

Forty years of software innovation have generated massive benefits for software creators, consumers and the economy. At the same time, the U.S. copyright regime has proved remarkably adept at balancing the interests of creators, consumers, and the public—including with regard to software embedded in consumer products. The coming decades will see a further burst of innovation, as the software and device industries reinvent themselves in an age of cloud computing. It is therefore critical that the Copyright Office tread with care so as not to alter the incentives or disturb the carefully developed legal precedent for software embedded in consumer products.

I. Software innovation is thriving

The pace and breadth of software innovation since Congress passed the 1976 Copyright Act have been astounding—and arguably unrivaled in the history of technology. At that time, most computers were large mainframes that primarily ran customized software. Given their massive size, cost, and complexity, they were used almost exclusively by large enterprises.³ But several key innovations in software and hardware quickly transformed the industry and made computing far more affordable, accessible, and powerful, such that today, software is nearly ubiquitous.

The first major stage in software’s post-1976 evolution was the advent of personal computers in the 1980s, enabled both by small, powerful microprocessors and flexible, easy-to-use operating systems and applications. Next, innovations in client-server software brought the power of networked computing to millions of small and medium-sized enterprises, spurring tremendous productivity gains and enabling them to compete more effectively with their larger rivals. Soon thereafter, the World Wide Web and the spread of broadband networks launched a new era of online computing as consumers and businesses leveraged the power of the Internet to deepen collaboration, speed innovation, and launch entirely new businesses and business models.

Today, we are in the midst of a transformational new generation of software innovation. Smartphones, tablets, and other mobile devices not only provide computing “on the go,” they also are bringing billions of new users online. Cloud computing is perhaps even more transformational: even as it powers the hundreds of thousands of apps that make mobile devices useful, it also makes truly hyperscale computing available to all. The numbers bear this out. Annual business spending on cloud services is estimated to exceed \$100 billion by

³ See, e.g., Bradford L. Smith, *The Third Industrial Revolution: Policymaking for the Internet*, 3 COLUM. SCI. & TECH. L. REV. 1, 19 (2001).

2017.⁴ By the end of 2016, consumers are likely to store more than 36 percent of their content in the cloud, including data, video, and backups from smartphones, tablet computers, and other mobile devices.⁵ The speed with which consumers have adopted cloud-capable devices like smartphones has been dramatic, with 64 percent of American adults owning smartphones in 2015, up from 35 percent in 2011.⁶ At the same time, advances in sensors and data analytics are ushering in an entire new generation of always-connected, “Internet of Things” (“IoT”) devices. Indeed, the number of these devices is expected to exceed 6 billion this year, a 30 percent increase over 2015.⁷

Microsoft is proud to have been at the forefront of each of these stages of software innovation. When Bill Gates and Paul Allen founded Microsoft in 1975, they were motivated by a bold vision: “a computer on every desk and in every home.”⁸ They sought to realize this vision by providing an open platform that enabled others to build compatible hardware and software, thus driving down costs and fueling innovation by thousands of software developers, device manufacturers, and service providers. As computing moved to new devices, Microsoft products like Visual Basic and services like Microsoft Store made available the creation and delivery of large-scale, easily created and distributed software and apps, democratizing computing for both developers and consumers. Microsoft was also one of the first major packaged software developers to make its software widely available for download, and to offer innovative software

⁴ See Brandon Butler, *IDC: Cloud will be \$107B industry by 2017*, TECHWORLD (Mar. 20, 2014), at <http://www.techworld.com/news/cloud/idc-cloud-will-be-107b-industry-by-2017-3507600/> (last accessed Feb. 12, 2016).

⁵ Gartner, *Gartner Says That Consumers Will Store More Than a Third of Their Digital Content in the Cloud by 2016* (June 25, 2012), at <http://www.gartner.com/newsroom/id/2060215> (last accessed Feb. 12, 2016).

⁶ Pew Research Center, *U.S. Smartphone Use in 2015* (Apr. 1, 2015), at <http://www.pewinternet.org/2015/04/01/us-smartphone-use-in-2015/> (last accessed Feb. 12, 2016).

⁷ See Gartner, *Gartner Says 6.4 Billion Connected “Things” Will Be in Use in 2016, Up 30 Percent From 2015* (Nov. 10, 2015), at <http://www.gartner.com/newsroom/id/3165317> (last accessed Feb. 12, 2016).

⁸ Microsoft, *Creating a Vibrant Information Technology Sector: Growth, Opportunity and Partnership* (Apr. 2002), at http://download.microsoft.com/documents/australia/corporateaffairs/opportunitieswhite_paper.pdf (last accessed Feb. 12, 2016).

functionality online as a service. Indeed, Microsoft began offering online services already in 1994—well before the term “cloud computing” had become common parlance.

Today, Microsoft continues to drive the next era of software innovation, with the goal of providing products and services that empower every person and organization on the planet to be more productive. More than 1.2 billion people around the world, representing 140 countries and 107 languages, create and share ideas with Microsoft’s Office software.⁹ Customers using our Azure cloud services can choose from a wide range of operating systems, including over 1000 that are open-source, to reduce costs, improve efficiency, and innovate.¹⁰ Through our CityNext and other initiatives, we are helping governments and enterprises around the world harness connected devices, cloud computing, and data analytics to improve their own operations and the services they provide to their customers and constituents.¹¹ And by making many of our most popular products and services available on multiple platforms (including Android and Apple’s iOS), we are expanding consumer choice and giving users greater flexibility and more options.¹²

As these examples illustrate, software rarely exists in isolation. Rather, it typically forms an integral part of a much larger ecosystem. For software embedded in consumer products, this ecosystem includes general-purpose software, apps, cloud-based software, and online services (including the enterprise software that runs such services), all of which would be affected by any change to the copyright regime for embedded software. For instance, over the next five year, 33 percent of consumers plan to adopt wearable fitness devices, and 25 percent are likely to adopt smart watches.¹³ Nearly all of these devices will rely on cloud

⁹ Microsoft, *By the Numbers*, at <http://news.microsoft.com/bythenumbers/planet-office> (last accessed Feb. 12, 2016).

¹⁰ Microsoft, *Linux in the Cloud*, at <https://www.microsoft.com/en-us/Openness/Linux> (last accessed Feb. 12, 2016).

¹¹ Microsoft, *Microsoft CityNext: Coming to a City Near You* (July 10, 2013), at <https://news.microsoft.com/2013/07/10/microsoft-citynext-coming-to-a-city-near-you/> (last accessed Feb. 12, 2016).

¹² Microsoft, *Microsoft Empowers Windows, iOS, Android, Mac, and Linux Developers to Reach Billions of New Customers* (Apr. 29, 2015), at <https://news.microsoft.com/2015/04/29/microsoft-empowers-windows-ios-android-mac-and-linux-developers-to-reach-billions-of-new-customers/%20> (last accessed Feb. 12, 2016).

¹³ Accenture, *The Internet of Things: The Future of Consumer Adoption* (2014), at <https://www.accenture.com/us-en/insight-internet-things-future-consumer-adoption.aspx> (last accessed Feb. 12, 2016).

services powered by enterprise software and will be deeply integrated with software on smartphones, tablets, and many other devices. Thus, seeking to draw artificial lines around software embedded in consumer products could have substantial unintended consequences: it will be difficult to apply in practice and could adversely affect incentives, investment, and innovation across many different sectors of the economy.

II. The U.S. copyright regime is remarkably effective in maintaining balance

Throughout these eras of software innovation, the U.S. copyright regime has proven remarkably effective in balancing incentives for creativity with the interests of users and the public. When debating the 1976 Act, Congress—recognizing the potential benefits of widespread computing—engaged in a thorough study of emerging uses of computers. Based on that study, it determined that it could best balance all relevant interests by codifying copyright protection for software while also making this protection subject to important exceptions.¹⁴

That decision proved prescient, as software developers have relied on copyright protection for over four decades to justify massive investments in software innovation. Microsoft was one of the first: in a 1976 “Open Letter to Hobbyists,” Bill Gates urged users to respect copyright in software and explained why such protection was critical to the software industry’s future development.¹⁵ As computer use became more widespread and software distribution methods evolved, developers continued to rely on copyright, first with shrink-wrap licenses (for packaged software products sold at retail) and later, click-wrap licenses (for software delivered online). Modern online services like streaming, online apps, and cloud computing rely on many of these same contractual and copyright concepts, enforced through end-user licenses and embodied in terms of service.¹⁶

¹⁴ H.R. Rep. 94-1476, at 1116 (1976).

¹⁵ Bill Gates, *An Open Letter to Hobbyists* (first published Feb. 3, 1976, in COMPUTER NOTES), at http://www.microsoft.com/about/companyinformation/timeline/timeline/docs/di_hobbyists.doc (last accessed Feb. 12, 2016).

¹⁶ See, e.g., Robert W. Gomulkiewicz, *The Federal Circuit’s Licensing Law Jurisprudence: Its Nature and Influence*, 84 WASH. L. REV. 199, 208 (2009) (“End-user licensing also allows software developers to offer packages of software and services; flexible client-server computing-usage models; and the same code to business users at one price, home users for a lower price, academic users for yet a lower price, and charitable organizations for free. In sum, end-user licensing is a key component of business-model innovation in the information economy.”).

The U.S. software industry, on the foundation of copyright protection, has grown into one of the most vibrant sectors of the economy. Between 1997 and 2012, annual U.S. software industry revenues grew from \$149 billion to \$425 billion.¹⁷ Without the protection of copyright law, investments in software during this period might well have stagnated, leaving consumers with fewer choices and less innovation.

In every step of this evolution, the U.S. copyright law has adroitly balanced the interests of software creators with those of users and the public. The Copyright Act itself has numerous “safety valves” that help to preserve this balance. One such safety valve is Section 107, codifying the fair use exception. Courts routinely apply fair use to protect the legitimate interests of software creators while also ensuring that users have the flexibility they need to use copyrighted software for new, transformative uses without being liable for copyright infringement.¹⁸ Another is in Section 109, which permits the resale of software embodied in devices and which cannot be copied during the ordinary operation or use of the device. Section 117 ensures that the basic operation of a computer—including a computer that runs the software in an everyday consumer product—does not put consumers at risk of copyright infringement. Perhaps most importantly, Section 1201 provides for rulemaking to mitigate the potential negative effects of the ban on the circumvention of technical protection measures for copyrighted software. For nearly two decades, Section 1201 rulemaking has functioned as a way to incrementally address and accommodate developing technologies affected by copyright based upon input from consumers, industry, and the general public.

In addition to these “safety valve” mechanisms in the Copyright Act itself, courts have applied the Act to new technologies and business models in ways that broadly help preserve the balance between the interests of creators, users, and the public. For example, in *Sony Corporation of America v. Universal City Studios*, the Supreme Court ruled that Sony’s manufacturing and sale of video cassette recorders that allowed consumers to watch television broadcasts on their own schedules—at the time a novel concept—was not copyright infringement.¹⁹ And in *Krause v. Titleserv, Inc.*, the Second

¹⁷ Software & Info. Industry Assoc., *The U.S. Software Industry: An Engine for Economic Growth and Employment* (2014), at http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2541673%20 (last accessed Feb. 12, 2016).

¹⁸ E.g., *Sega Enters. Ltd. v. Accolade, Inc.*, 977 F.2d 1510, 1527-28 (9th Cir. 1992) (ruling that reverse-engineering copyrighted code in a video game console in order access the code’s unprotectable functionality was fair use).

¹⁹ 464 U.S. 417 (1984).

Circuit affirmed that Section 117 allows consumers to modify their copies of software programs in order to correct bugs or add routine functionality needed to ensure that those programs continue to operate and provide value.²⁰

U.S. leadership on robust yet balanced copyright laws has made copyright protection for software the global norm, providing substantial benefits to U.S. industry and the economy. These efforts have been critical to harmonizing international law and practice, including through agreements such as the WTO Agreement on Trade-Related Aspects of Intellectual Property Rights, the WIPO Copyright Treaties, and numerous U.S. free trade agreements.²¹ Our Nation's commitment to this vital balance is reflected most recently in the proposed Trans-Pacific Partnership Agreement ("TPP"), which requires TPP Parties to provide strong copyright protection while "endeavor[ing] to achieve an appropriate balance in its copyright and related rights system."²² Remaining steadfast in its commitment to these principles in trade agreements is critical to maintaining U.S. leadership in technology and software innovation.

The U.S. copyright regime has helped software innovation and choice flourish—and consumers are the winners. Software developers rely on copyright for a wide array of licensing and distribution models. This includes not only proprietary software, but also a range of open-source software development and licensing models. For example, the differences among various PC, mobile device, and mobile app ecosystems run the gamut from open-source-based development and licensing approaches²³ to more proprietary or curated approaches, each of which strikes a different balance between variation and fragmentation. The success of all of these approaches, as evidenced in Google's Android, Apple's iOS, and Microsoft's Store, and myriad devices in the marketplace using these platforms, demonstrates that different consumers want different things—and that software developers, operating on a foundation of copyright even as they pursue different

²⁰ 402 F.3d 119 (2d Cir. 2005).

²¹ WTO Agreement on Trade-related Aspects of Intellectual Property Rights (Apr. 15, 1994), at https://www.wto.org/english/tratop_e/trips_e/t_agm0_e.htm (last accessed Feb. 12, 2016); WIPO Copyright Treaty (Dec. 20, 1996.), at http://www.wipo.int/treaties/en/text.jsp?file_id=295166 (last accessed Feb. 12, 2016).

²² Trans-Pacific Partnership Agreement, art. 18.66, at <https://medium.com/the-trans-pacific-partnership/intellectual-property-3479efdc7adf> (last accessed Feb. 12, 2016).

²³ Indeed, copyright law underpins the open-source licensing, development, and distribution model and is critical to enforcing such licenses. *See, e.g., Jacobsen v. Katzer*, 535 F.3d 1373, 1382 (Fed. Cir. 2008).

aims, are serving those various needs exceedingly well. The genius of the existing U.S. copyright regime is that it has proven sufficiently flexible and balanced to provide the legal basis for all of these different approaches.

This flexibility is reflected in Microsoft's own offerings as well. Users of our Azure cloud services can run a wide range of operating systems and platforms, and a significant percentage of Azure customers use applications that run on open-source operating systems. Indeed, more than 20 percent of the workloads running on Azure today run a version of Linux. The choice remains entirely with the user. Far from restricting such choice, copyright protection for software has helped enable this choice by giving both Microsoft and its customers the freedom and flexibility choose whatever products, services, and business models best fit their needs.

In sum, copyright protection for software has generated enormous innovation while successfully balancing the interests of creators with those of users and the public. It underpins a wide array of software products, businesses, and business models, giving consumers a tremendous amount of choice while spurring competition. This is no less true with regard to software embedded in consumer products, an area in which software innovation and choice are clearly flourishing.

III. The current system's success recommends a cautious approach to legislation and policymaking

Against this backdrop, the Copyright Office should proceed with caution before recommending changes that would alter copyright protection for software embedded in consumer products. As discussed, the evidence is overwhelming that software innovation—including in embedded software—is flourishing and that consumers and the public are benefitting enormously. By contrast, there is remarkably little evidence that copyright protection for software embedded in consumer products is causing any systemic problems. Indeed, when the Department of Commerce's Internet Policy Task Force recently completed its 2016 study on novel copyright issues in the digital economy, it found the U.S. copyright regime to be sufficiently flexible to adapt to the evolution of software in consumer products, and therefore did not recommend any change to existing copyright protections for such software.²⁴ In this context, there is no reason to believe

²⁴ Dep't of Commerce, Internet Policy Task Force, *White Paper on Remixes, First Sale, and Statutory Damages*, 64 (2016), at <http://www.uspto.gov/sites/default/files/documents/copyrightwhitepaper.pdf> (last accessed Feb. 12, 2016).

that a different regime would be more effective in spurring innovation or at balancing the interests of creators, users, and the public.

This is not to say that tensions in the system never arise. But when they do, existing provisions in the Copyright Act, combined with agency rulemaking, judicial interpretations, and voluntary private-sector efforts, have proven up to the task of maintaining the right balance. For example, in 2015, after considering the input of industry, public interest groups, and private citizens, the Library of Congress, acting pursuant to Section 1201, adopted exemptions for circumventing copyright protection on computer systems in vehicles for the purpose of diagnosis, repair, or lawful modification.²⁵

Accordingly, we urge the Copyright Office not to recommend changes to copyright protection for software embedded in consumer products at this time. Businesses in numerous sectors of our economy today rely on the certainty that the existing copyright system provides for embedded software. Even the *potential* for changes to that system could deter investment and innovation, as inventors and companies seek to avoid even the risk that they may be unable to recoup their investments in software development.

Sincerely,



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²⁵ Exemption to Prohibition on Circumvention of Copyright Protection Systems for Access Control Technologies, 80 Fed. Reg. 65944, 95654-55 (Library of Congress, Oct. 28, 2015).